“Fake News Detection project”

A report submitted in partial fulfillment of the Academic requirements for the award of the degree of

Bachelor of Technology

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UNDER THE COURSE

MACHINE LEARNING LABORATORY

CMR COLLEGE OF ENGINEERING & TECHNOLOGY
(Autonomous)
(NAAC Accredited with ‘A^{+}’ Grade & NBA Accredited)
(Approved by AICTE, Permanently Affiliated to JNTU Hyderabad)
KANDLAKOYA, MEDCHAL ROAD, HYDERABAD-501401

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CERTIFICATE

This is to certify that the report entitled **"Fake News Detection project"** is a bonafide work done by **HARSHA DHANUSH YETURI -23H51A0525, JASHWANTH JUMPALA-23H51A0528, PANJALA PAVAN -23H51A0549 .** In partial fulfillment of the requirement for completion of the "**MACHINE LEARNING LABORATORY**" of III-B-Tech II- Semester is a record of a bonafide work carried out under guidance and supervision.

Signature of Faculty **Signature of HOD**

DECLARATION

We are obliged and grateful to thank, CMRCET, for his cooperation in all respects during the course.

We would like to thank the Principal of CMRCET, Dr.A. Sheshu Kumar, for his support in the course of this project work. Finally, we thank all our faculty members and Lab Programmer for their valid support.

We own all our success to our beloved parents, whose vision, love and inspiration has made us reach out for these glories.

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CHAPTER 1

ABSTRACT

Fake news detection is the process of identifying false or misleading news articles using machine learning techniques. With the rapid growth of social media platforms like Facebook and Twitter, fake news spreads quickly and influences public opinion. This project uses Natural Language Processing (NLP) and machine learning algorithms to classify news as real or fake. The dataset is preprocessed, converted into numerical form using TF-IDF vectorization, and trained using a classifier like Logistic Regression. The model achieves high accuracy in detecting fake news articles.

CHAPTER 2

INTRODUCTION

Fake news means false or misleading information that is spread as real news. Today, social media platforms like Facebook and Twitter help news spread very fast. Because of this, fake news also spreads quickly and can create confusion and problems in society. To solve this issue, Machine Learning is used to automatically detect whether a news article is real or fake. By analyzing the text of the news using techniques like Natural Language Processing (NLP), the system can classify the news correctly. This project aims to build a simple and accurate model to identify fake news.

Additionally, fake news can influence people's opinions and decisions in areas such as politics, health, and public safety. Detecting fake news manually is difficult because of the huge amount of information shared online every day. Machine learning models can analyze patterns in text data and learn from large datasets. Techniques like TF-IDF and classification algorithms help the system understand important words and features in the news. This helps in improving the accuracy and reliability of fake news detection systems.

CHAPTER 3

PROBLEM STATEMENT:

The major problem is the rapid spread of fake news through social media and online platforms. It creates misinformation, panic, and political manipulation.This project uses Natural Language Processing (NLP) and machine learning algorithms to classify news as real or fake. The dataset is preprocessed, converted into numerical form using TF-IDF vectorization, and trained using a classifier like Logistic Regression. The model achieves high accuracy in detecting fake news articles.

Key Points of the Problem Statement

Rapid spread of fake news through social media and online platforms.

Causes misinformation, panic, and political manipulation.

Uses **Natural Language Processing (NLP)** and **Machine Learning**.

News text converted into numerical form using **TF-IDF Vectorization**.

Model trained using **Logistic Regression classifier**.

Achieves **high accuracy in detecting fake news articles**.

METHODOLOGY

The methodology for fake news detection begins with collecting the dataset containing fake and real news articles. The dataset is then preprocessed by cleaning the text, removing stopwords, punctuation, and unnecessary characters to improve the quality of the data. After preprocessing, the news articles are labeled as fake or real to prepare them for training the machine learning model. The textual data is then converted into numerical format using TF-IDF (Term Frequency–Inverse Document Frequency) vectorization so that the model can understand the importance of words in the documents.

Next, the dataset is divided into training and testing data. A machine learning algorithm such as Logistic Regression is used to train the model on the training data. After training, the model is tested using the test dataset to evaluate its performance. Finally, the model's performance is measured using evaluation metrics such as Accuracy, Precision, Recall, and F1-Score to determine how effectively the system can detect fake news.

Methodology Steps

The methodology for Fake News Detection includes the following steps:

Data Collection

Collect a dataset containing real and fake news articles.

2. Data Preprocessing

Clean the text by removing punctuation, stopwords, and converting text to lowercase.

3. Feature Extraction

Convert text data into numerical form using techniques like TF-IDF.

4. Model Training

Train a machine learning algorithm such as Logistic Regression using the training dataset.

5. Model Testing

Test the model using test data to check its performance.

6. Evaluation

Measure accuracy, precision, recall, and F1-score to evaluate the model performance

MODEL TRAINING

Model training using:Logistic regression

Split the dataset into **training set and testing set**.

Convert the news text into numerical features using **TF-IDF Vectorization**.

Apply the **Logistic Regression algorithm** to train the model.

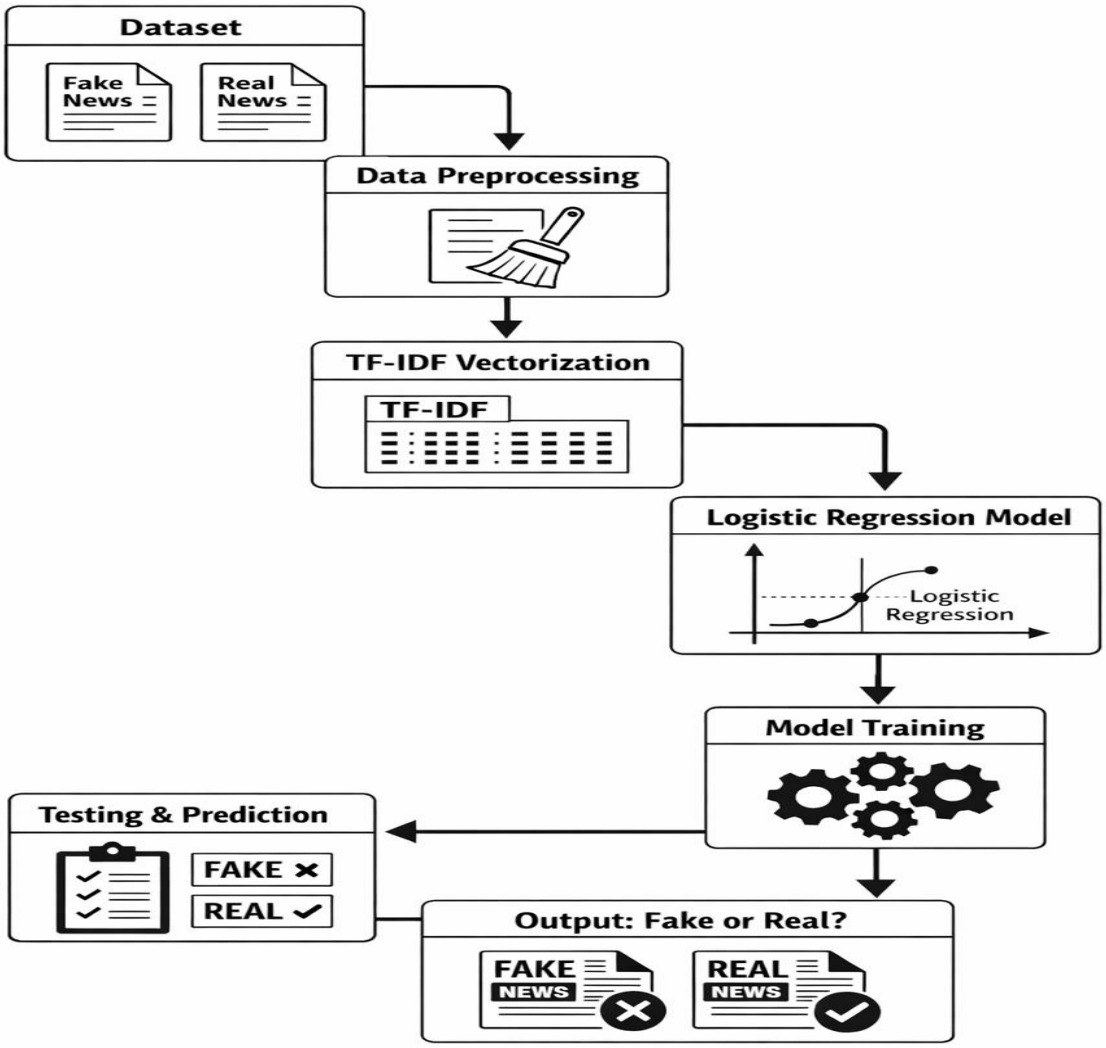
The model learns the relationship between **news text features and labels (Fake or Real)**.

Training is performed using the **training dataset**.

After training, the model can **predict whether a given news article is fake or real**.

CHAPTER 4

FLOWCHART



CHAPTER 5

SOURCE CODE

```
import pandas as pd
import string
import re

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# STEP 1: Load datasets
fake = pd.read_csv("Fake.csv")
real = pd.read_csv("True.csv")

# STEP 2: Add labels
fake["label"] = 0  # Fake news
real["label"] = 1   # Real news

# STEP 3: Combine datasets
data = pd.concat([fake, real])
# STEP 4: Keep only text and label columns
data = data[["text", "label"]
```

```
# STEP 5: Clean text def clean_text(text):
        text = str(text).lower()

        text = re.sub(r'[%s]' % re.escape(string.punctuation), '', text)

        return text


    data["text"] = data["text"].apply(clean_text)


    # STEP 6: Split dataset

    X = data["text"]

    y = data["label"]


    X_train, X_test, y_train, y_test = train_test_split(

        X, y, test_size=0.25, random_state=42)


    # STEP 7: Convert text into numbers (TF-IDF)

    vectorizer = TfidfVectorizer()

    Xv_train = vectorizer.fit_transform(X_train)

    Xv_test = vectorizer.transform(X_test)


    # STEP 8: Train model

    model = LogisticRegression(max_iter=1000)
```

model.fit(Xv_train, y_train)

OUTPUT:

```
C:\Users\harsh\OneDrive\Desktop\harsha>python fake_news_detection.py
Final Accuracy: 0.9864587973273942

Confusion Matrix:
 [[5809   86]
 [  66 5264]]

Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.99      0.99      5895
           1       0.98      0.99      0.99      5330

    accuracy                           0.99     11225
   macro avg       0.99      0.99      0.99     11225
weighted avg       0.99      0.99      0.99     11225


C:\Users\harsh\OneDrive\Desktop\harsha>
```

CHAPTER 6

RESULTS AND DISCUSSION

Results:

The fake news detection model was successfully implemented using Machine Learning with Logistic Regression. After training and testing the model on the dataset, the system achieved an accuracy of 98.64%, which indicates that the model can correctly classify most news articles as fake or real.

The confusion matrix shows that the model correctly predicted 5809 fake news articles and 5264 real news articles, with only a small number of misclassifications. This demonstrates that the model performs very well in distinguishing between fake and real news.

The classification report also shows high values for precision, recall, and F1-score (around 0.99) for both classes, indicating strong performance and reliability of the model.

Discussion:

The high accuracy of the model indicates that **TF-IDF vectorization combined with Logistic Regression** is effective for text classification problems like fake news detection. The model successfully learns important patterns and keywords present in fake and real news articles.

However, the performance of the system mainly depends on the quality and size of the dataset. If the dataset changes or contains new types of news articles, the model may need retraining to maintain high accuracy. Future improvements can include using advanced deep learning models and larger datasets to further enhance the detection capability.

✅**Final Accuracy Achieved:** 98.64%
✅**Evaluation Metrics Used:** Accuracy, Precision, Recall, F1-score, Confusion Matrix

CHAPTER 7

CONCLUSION

The Fake News Detection system successfully classifies news articles using Logistic Regression and TF-IDF vectorization. The model achieves high accuracy and can be used to reduce the spread of misinformation on online platforms. Machine learning proves to be effective for text classification problems by identifying patterns in news content. This system helps in automatically detecting fake news and improving the reliability of information shared on the internet. In the future, the system can be enhanced by using larger datasets and advanced deep learning techniques to further improve its accuracy and performance.

CHAPTER 8

Future Enhancement

Future improvements of the fake news detection system can include the use of advanced deep learning models such as LSTM or BERT to achieve higher accuracy in classification. The system can also be developed into a real-time fake news detection system that can analyze news articles instantly. Integration with social media platforms can help identify and prevent the rapid spread of fake news online. Additionally, a browser extension can be created to automatically alert users when they encounter suspicious or fake news content. The system can also be improved by using multilingual datasets to detect fake news in different regional languages.

CHAPTER 9

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